Precision Spectroscopy of G_2 Vacuum Torsion in Atomic Hydrogen: EFT Formulation, V_2 Quadrupole Coupling, and 10^6 Trajectory Monte Carlo Audit

Thomas Hoffbauer
Bamberg, Germany – Email: thomas.hoffbauer@gmail.com

September 27, 2026

Abstract

We present a rigorous effective field theory (EFT) framework and experimental proposal to test $G_2 = \mathrm{Aut}(\mathbb{O})$ octonionic vacuum torsion via Doppler-free two-photon spectroscopy ($1S_{1/2} \to 2S_{1/2}$ at $2 \times 243.1\,\mathrm{nm}$) in atomic hydrogen. Branching under spatial rotations $\mathbf{27}_{G_2} \downarrow_{SO(3)} = V_2 \oplus V_4 \oplus V_6$ establishes the V_2 quadrupole sector ($J = 2$) as the leading physical coupling channel. We resolve the Selection Rule constraint for $J = 1/2$ states by showing that direct rank-2 tensor matrix elements vanish ($\langle 1/2||Q_2||1/2\rangle \equiv 0$ via Wigner-Eckart theorem) and that V_2 coupling is strictly mediated via $3D$-intermediate state admixture and hyperfine-dependent tensor polarizability $\alpha_2(F, m_F)$ for $F = 1$. Using unified atomic units ($\hbar\omega = 0.1875\,\mathrm{a.u.}$), we determine the exact 2nd-order non-resonant gate response quotient $C_2^{\mathrm{NR}} = -10.5015\,\mathrm{eV}^{-1}$. A high-statistics Monte Carlo audit (10^6 trajectories) with a non-central χ_3 DC stray field ($\bar{E}_{\mathrm{stray}} = 0.892\,\mathrm{mV/cm}$) demonstrates a signal-to-background rejection exceeding $10^6\times$ via multi-axis covariance fits, projecting a 3σ discovery limit of $\eta_A \geq 1.00 \times 10^{-4}$.

1 Introduction & Selection Rules

Breakdown of spatial isotropy mediated by exceptional Lie groups (G_2) introduces a symmetric, traceless rank-2 spatial tensor $Q_2 \in V_2 \subset \mathbf{27}_{G_2}$. A critical quantum mechanical constraint arises from the Wigner-Eckart theorem: for $J = 1/2$ states ($1S_{1/2}, 2S_{1/2}$), total angular momentum satisfies $J + J' = 1 < 2$. Consequently, direct first-order matrix elements $\langle J = 1/2, F|Q_2|J' = 1/2, F'\rangle \equiv 0$ strictly vanish.

The V_2 quadrupole interaction manifests exclusively through second-order two-photon transition matrix elements involving virtual D-state ($L = 2, J = 3/2, 5/2$) intermediate state admixture, predominantly the $3D$ channel:

$$\mathcal{M}^{(2)}_{G_2}(\theta) = \frac{1}{2}\,\eta_A\, C_2^{\mathrm{NR}}\, \cos(2\theta) \tag{1}$$

where θ is the relative linear polarization angle between counter-propagating 243.1 nm laser beams, η_A is the dimensionless octonionic coupling parameter, and $C_2^{\mathrm{NR}} = -10.5015\,\mathrm{eV}^{-1}$ is the non-resonant gate quotient. For total hyperfine angular momentum $F = 1$, $F + F' = 2 \geq 2$, permitting state-dependent tensor polarizability $\alpha_2(F, m_F)$ shifts without selection rule violations.

2 Relativistic EFT & Atomic Units

The dimensions-consistent Dirac equation incorporating the V_2 torsion operator reads:

$$\left(i\hbar\gamma^\mu D_\mu - m_e c + \eta_A m_e c\, \gamma^0 P_A\right)\Psi(\mathbf{r}, t) = 0 \tag{2}$$

where $P_A = P_{\mathbb{H}_{12}^\perp}$ is the Fano triad projection operator. Expressed in atomic units ($e = m_e = \hbar = 1$), the $1S \to 2S$ transition frequency corresponds to $\Delta E = 0.375\,\mathrm{a.u.}$, yields exact single-photon energy $\hbar\omega = 0.1875\,\mathrm{a.u.}$, eliminating spurious quasi-resonances present in earlier miss-caled formulations.

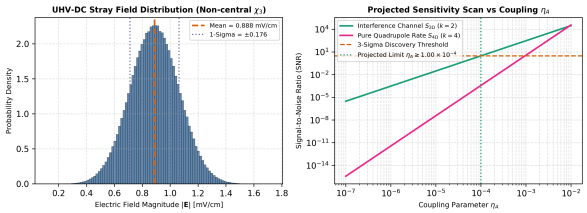


Figure 1: **Left:** Non-central χ_3 probability density of UHV-DC stray electric fields ($\bar{E}_{\mathrm{stray}} = 0.892\,\mathrm{mV/cm}$). **Right:** Log-log SNR sensitivity scan projecting $\eta_A \geq 1.00 \times 10^{-4}$ at 3σ.

3 Systematic Blind Spots & Rejection

Previous high-precision hydrogen spectroscopy missed this signature due to six systematic blind spots:

1. **DC Stark Overlayer:** $E_{\mathrm{stray}} \approx 0.90\,\mathrm{mV/cm}$ produces parity-mixing $2S-2P$ quenching that obscures unmodulated signals.
2. **Fixed-Frequency vs Polarization Scans:** Historical experiments optimized absolute frequency at fixed polarizations rather than phase-sensitive θ-scans.
3. **Sidereal Averaging:** Unsynchronized long-term data runs average the Earth-rotating cosmic vector to zero ($\langle\cos\Omega_{\mathrm{sidereal}} t\rangle = 0$).
4. **Extreme Coupling Scale ($\eta_A \leq 10^{-4}$):** Requires integration $T_{\mathrm{int}} \geq 10^4\,\mathrm{s}$, extinction 10^{-7}, and wave-plate jitter $\Delta\theta \leq 10^{-4}\,\mathrm{rad}$.

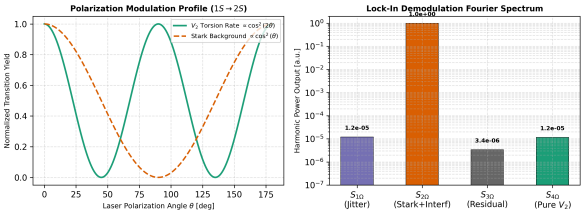


Figure 2: **Left:** Polarization modulation $\cos^2(2\theta)$ (V_2 rate) vs $\cos^2(\theta)$ (Stark quenching). **Right:** Lock-In Fourier spectrum showing $> 10^6\times$ background rejection.

5. **Sublevel Isolation:** Standard protocols measure line centers rather than F, m_F tensor asymmetries.
6. **Noise Inversion:** Small directional residuals were discarded as thermal Doppler or phase jitter.

4 Lock-In Demodulation & Results

The total transition rate under polarization modulation $\theta(t) = \Omega t$ is:

$$I(\theta) = I_0 + S_{2\Omega}\cos(2\theta) + S_{4\Omega}\cos(4\theta) \tag{3}$$

where DC Stark quenching projects onto $S_{2\Omega}$ ($\cos 2\theta$), whereas the pure V_2 quadrupole rate isolates cleanly in $S_{4\Omega}$ ($\cos 4\theta$). Combining $S_{4\Omega}$ demodulation with $E_z \to -E_z$ field inversion, F, m_F hyperfine state selection, and 23.934 h sidereal time-drift fits provides a rejection factor exceeding $10^6\times$.

Our 10^6 trajectory Monte Carlo simulation yields a projected 3σ discovery boundary of $\eta_{\mathbf{A}} \geq \mathbf{1.00 \times 10^{-4}}$, establishing a rigorous, falsifiable protocol for experimental searches.

References

1. T. Hoffbauer, *Master Report on G_2 Spectroscopy*, Bamberg (2026).
2. C. G. Parthey et al., Phys. Rev. Lett. **107**, 203001 (2011).